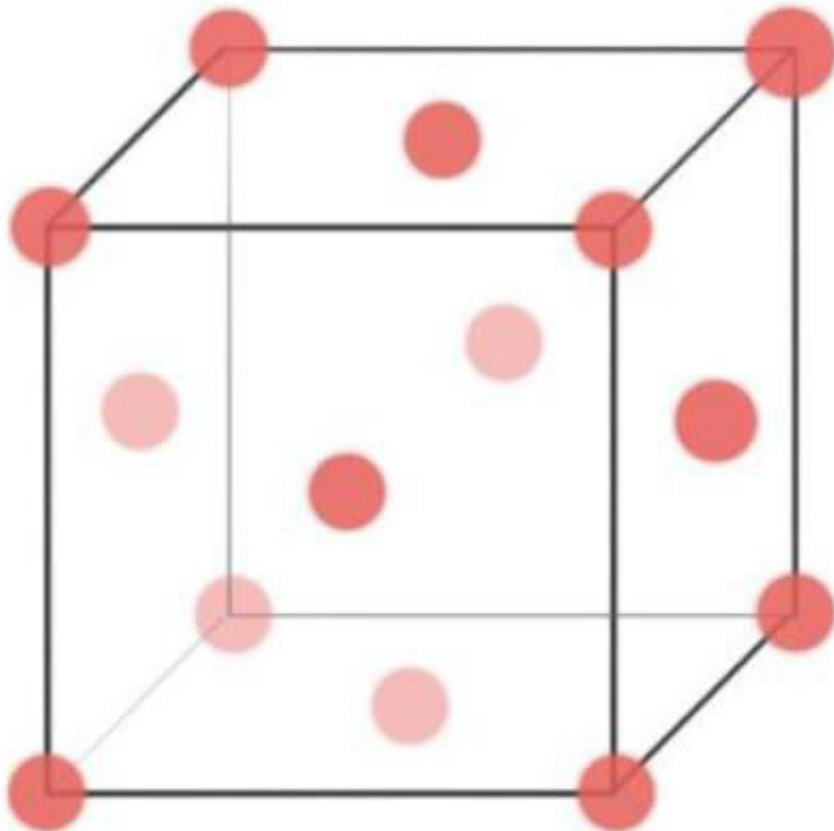


APS110 Tutorial #5

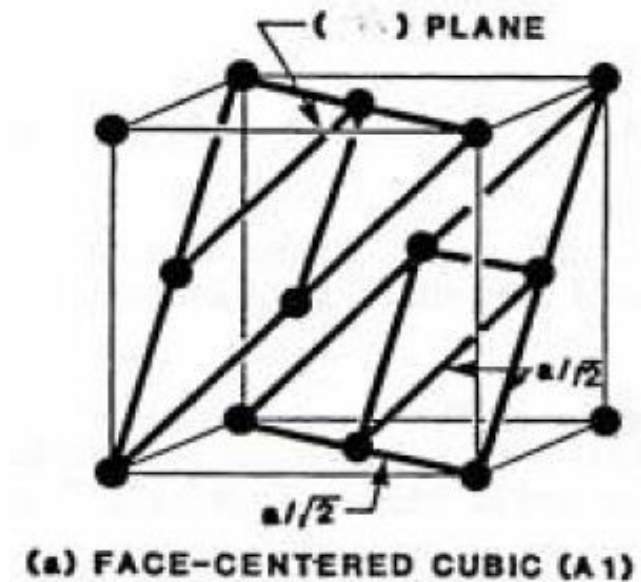
Daniel Brlas

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Review – FCC



- Coordination number is the number of nearest neighbour atoms
- In the FCC structure, the CN is 12
- The atomic packing factor for FCC is 0.74



Review – NaCl

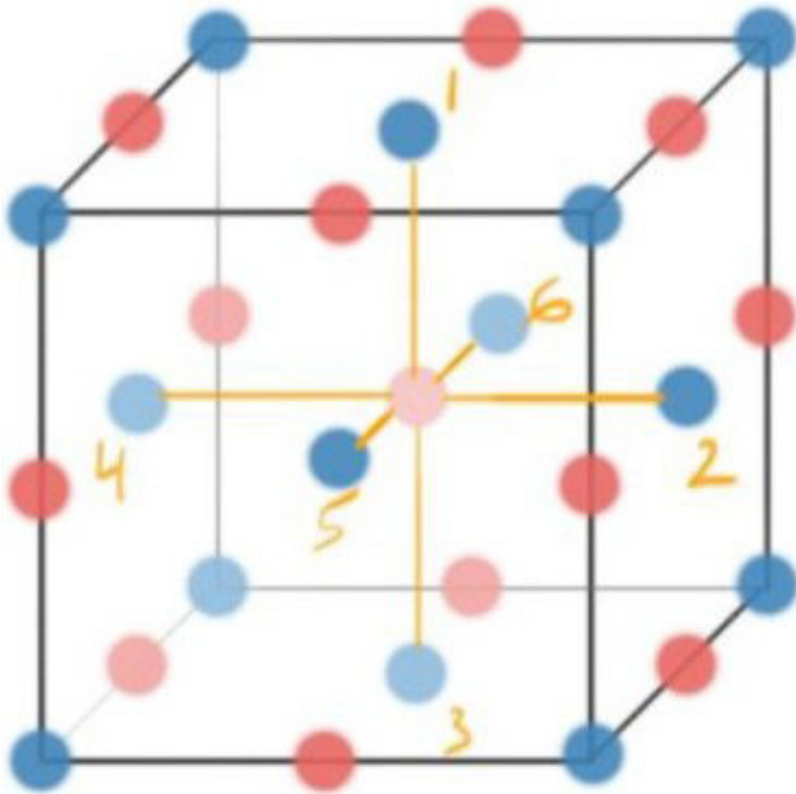


Figure 3: The rock salt ceramic crystal structure.

- In the NaCl structure, the CN between cations and anions is 6
- The atomic packing factor for the NaCl structure is 0.79 when the R_c/R_a ratio is 0.414

Review – BCC

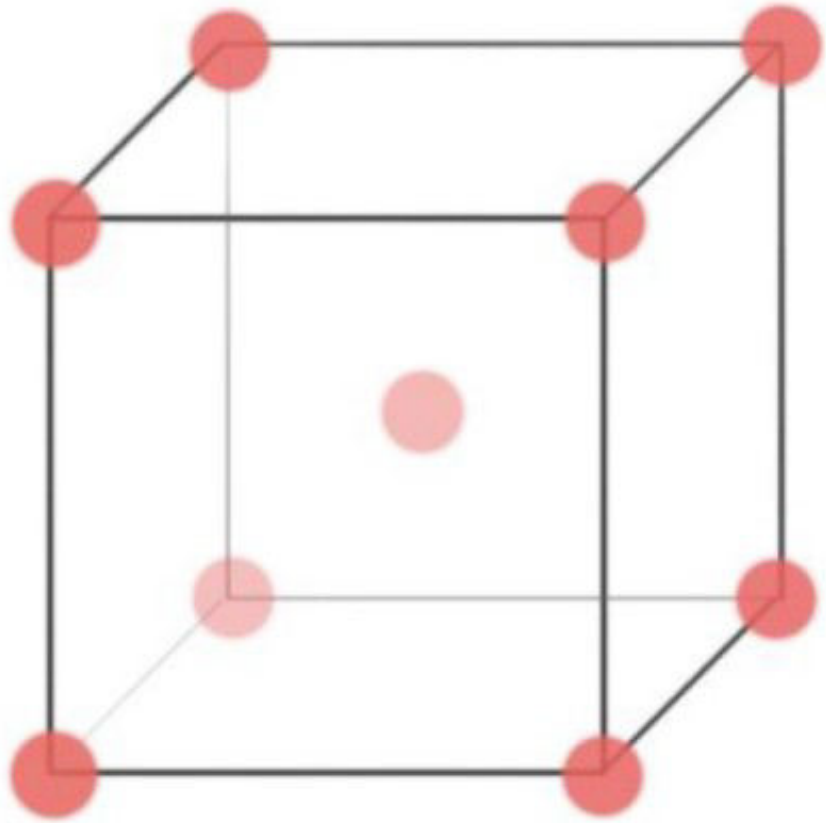
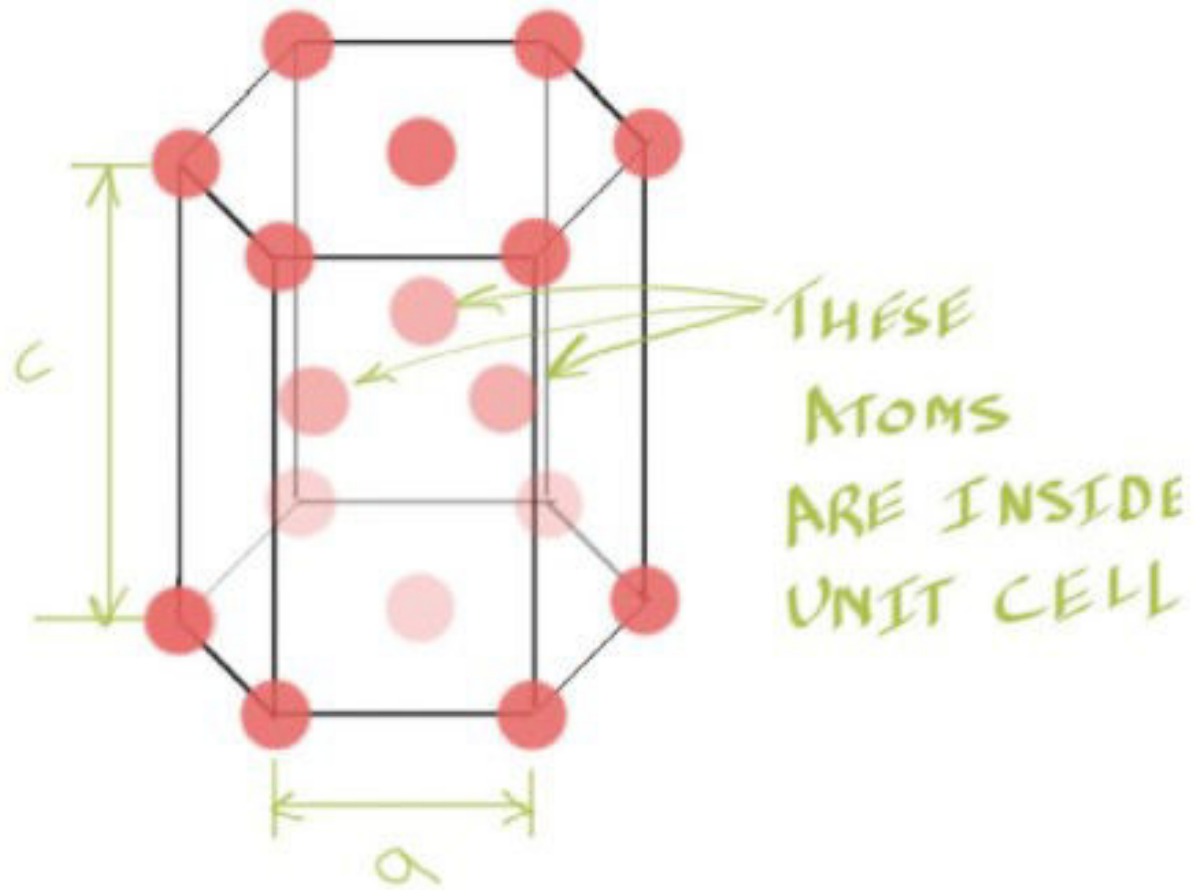


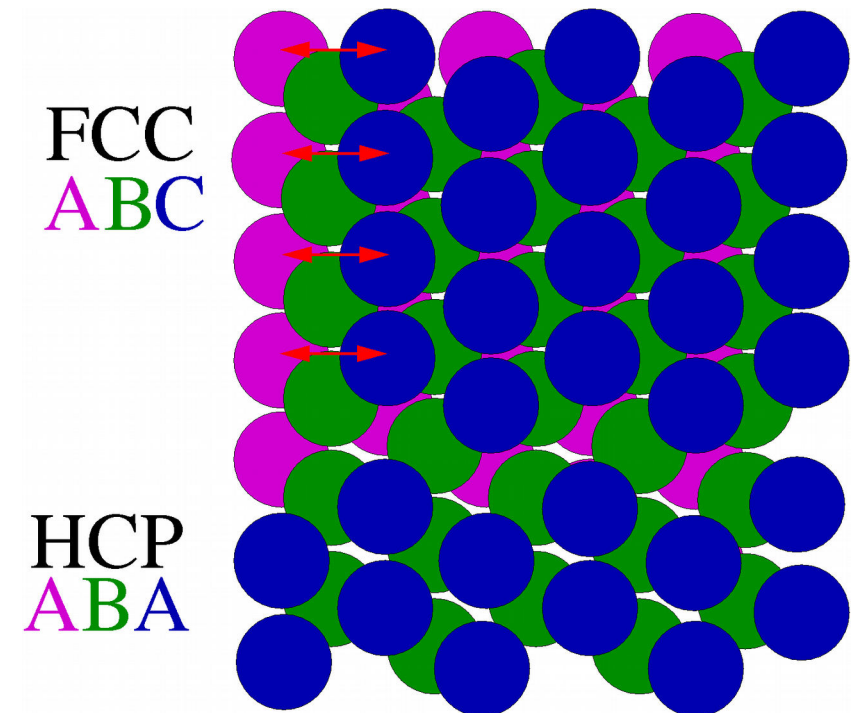
Figure 4: The body centred cubic crystal structure.

- In the BCC structure, the CN is 8
- The atomic packing factor for BCC is 0.68

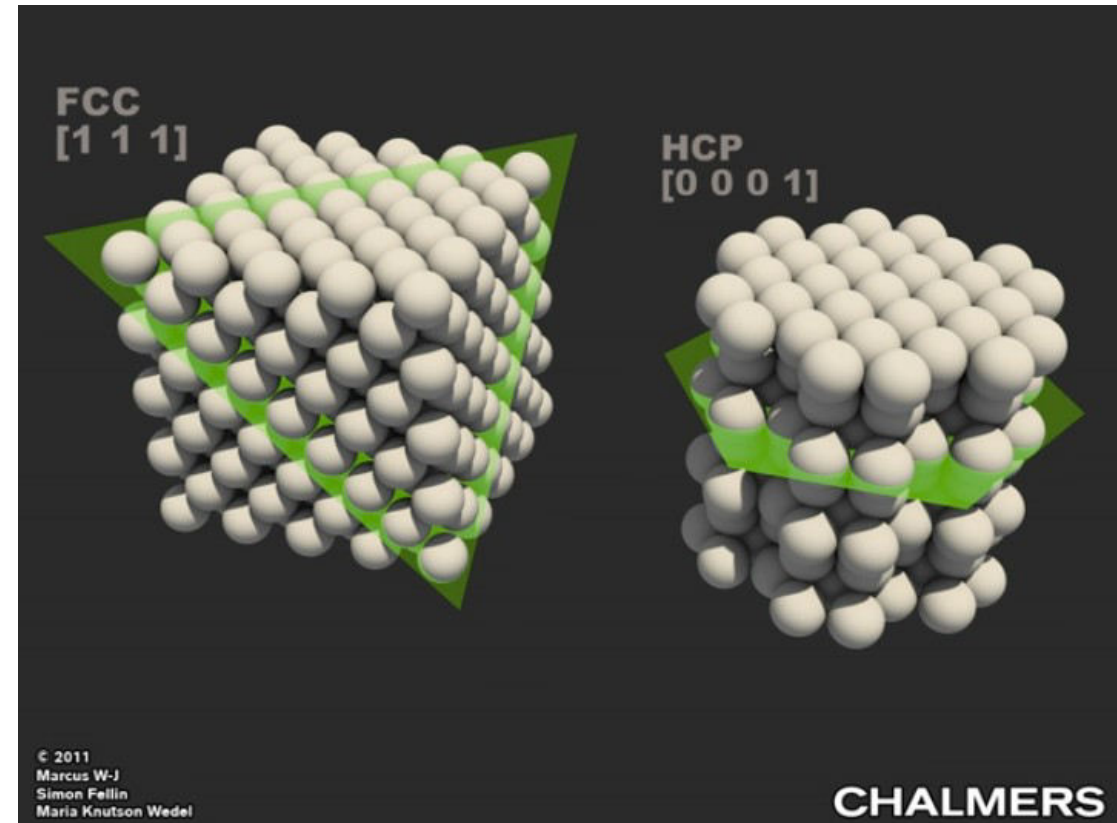
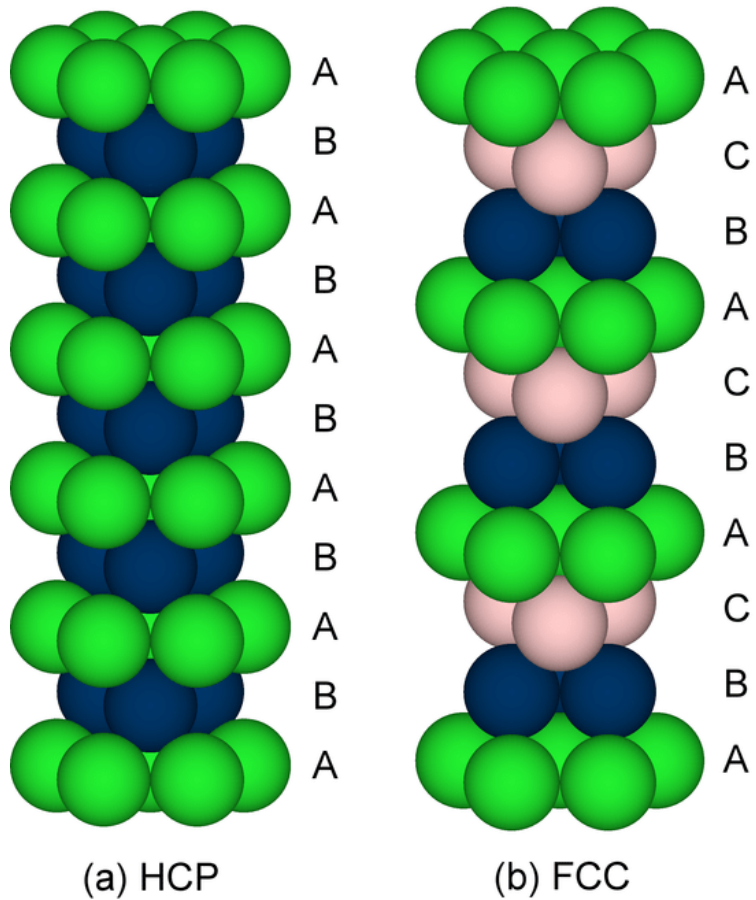
Review – HCP



- In the HCP structure, the CN is 12
- The atomic packing factor for HCP is 0.74



Review – FCC vs. HCP



Practice Question 1

1. Below are listed the atomic weight, density, and atomic radius for three hypothetical alloys. For each determine whether its crystal structure is FCC, BCC, or simple cubic and justify why.

Alloy	Atomic Weight (g mol ⁻¹)	Density (g cm ⁻³)	Atomic Radius (Å)
A	43.1	6.40	1.22
B	184.4	12.30	1.46
C	91.6	9.60	1.37

Practice Question 1

Assuming random crystal structures to compare (official solution):

$$\textcircled{A} \text{ Assume BCC } \rightarrow \rho = \frac{n \cdot A_a}{\left(\frac{4R}{\sqrt{3}}\right)^3 N_a} = \frac{(2 \cdot 43.1)}{\left(\frac{4 \cdot 1.22 \times 10^{-8} \text{ cm}}{\sqrt{3}}\right)^3 \cdot N_a}$$
$$= 6.40$$

$$\textcircled{B} \text{ Assume simple cubic } \rho = \frac{n \cdot A_B}{(2a)^3 N_a} = \frac{(1 \times 184.4)}{(2 \times 1.46 \times 10^{-8})^3 \cdot N_a}$$
$$= 12.3 \text{ g/cm}^3$$

$$\textcircled{C} \text{ BCC } \rho = \frac{n A_C}{\left(\frac{4R}{\sqrt{3}}\right)^3 N_a} = \frac{(2 \times 91.6)}{\left(\frac{4 \cdot 1.37 \times 10^{-8}}{\sqrt{3}}\right)^3 \cdot N_a}$$
$$= 9.6 \text{ g/cm}^3$$

Practice Question 1 (continued)

Isolating APF (your solution, legit impressed good job everyone):

$$\rho = \frac{nA}{V_c N_A} \text{ and } APF = \frac{V_a}{V_c}, \text{ where } V_a = n \cdot \frac{4}{3}\pi R^3$$

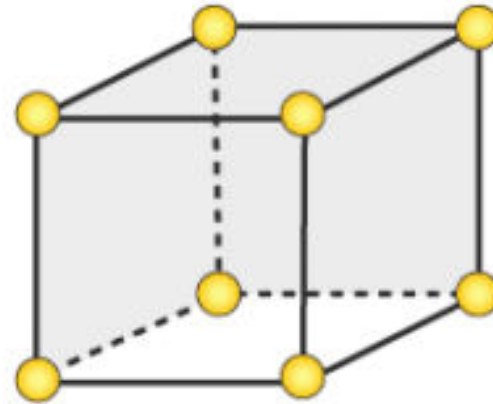
$$\rho = \frac{n}{V_c} \cdot \frac{\frac{4}{3}\pi R^3}{\frac{4}{3}\pi R^3} \cdot \frac{A}{N_A} = \frac{V_a}{V_c} \cdot \frac{A}{\frac{4}{3}\pi R^3 \cdot N_A}$$

$$\rho = APF \cdot \frac{3A}{4\pi R^3 \cdot N_A} \Rightarrow APF = \frac{4\pi R^3 \cdot N_A \cdot \rho}{3A}$$

Now just plug the numbers in and find out which crystal structure based on APF
(Note, doesn't work for HCP vs. FCC but you can't compare those based on density anyway)

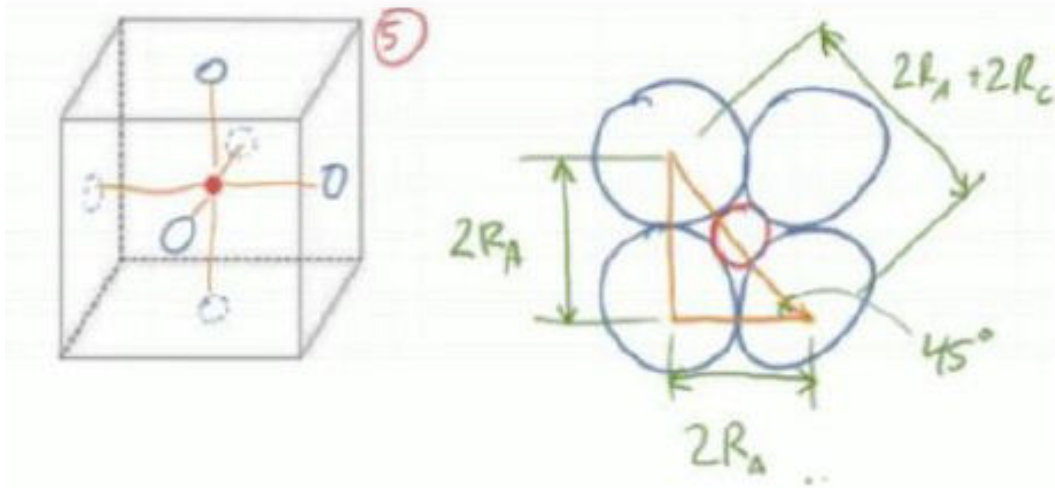
Practice Question 2

2. Demonstrate that the ideal cation-to-anion ratio (R_c/R_a) for a cation that just fits into the octahedral interstitial site is 0.414. Begin with a sketch of one octahedral site within the unit cell below, clearly showing the coordination number.



Practice Question 2

Same as Top Hat lecture notes:



$$\sin 45^\circ = \frac{2R_A}{2R_A + 2R_C}$$

$$R_A \sin 45^\circ + R_C \sin 45^\circ = R_A$$

$$\frac{R_A}{R_A} \sin 45^\circ = \frac{R_C}{R_A} \sin 45^\circ = \frac{R_A}{R_A}$$

$$\frac{R_C}{R_A} = \frac{1 - \sin 45^\circ}{\sin 45^\circ} = 0.414$$

Practice Quiz

1. Which of the following crystal structures contain sites with a coordination number of 8?
 - i. BCC
 - ii. FCC
 - iii. NaCl structure
 - iv. Simple cubic
 - v. Octahedral interstitial site
2. For a BCC crystal composed of a single element, what is the atomic radius of that element in terms of the lattice parameter a ?
3. A researcher has obtained the data in the table below from the internet. Unfortunately, the table is incomplete. If you don't help them fill out the missing data, indicated by the bold numbers, they might have to do actual experiments themselves.

Element/Compound	Atomic Radius (nm)	Crystal Structure	Molar mass (g mol ⁻¹)	Density (g cm ⁻³)
Cu	1	FCC	63.546	8.96
Co	0.200	HCP	58.933	4
Po	0.168	Simple Cubic	3	9.196
Sc	0.162	HCP	44.955	5

2.

$$APF = \frac{n \cdot \frac{4}{3}\pi R^3}{V_c}, V_c = a^3, n_{BCC} = 2$$

3. (Written in order going down the table): $a^2 + a^2 + a^2 = (4R)^2 \Rightarrow a = \frac{4}{\sqrt{3}}R$

3. (Written in order going down the table):

① for FCC Copper

$$\hookrightarrow \rho = \frac{n \cdot A}{a^3 \cdot N_A} \quad \left| \quad a = 2\sqrt{2}R \right.$$

$$\therefore \rho = \frac{n \cdot A}{(2\sqrt{2}R)^3 \cdot N_A} \Rightarrow R = \left(\frac{nA}{\rho N_A} \right)^{1/3} \cdot \frac{1}{2\sqrt{2}}$$

$$R = \left(\frac{4 \cdot 63.546}{8.96 \times 10^6 \cdot N_A} \right) \cdot \frac{1}{2\sqrt{2}} = 127 \text{ pm}$$

④ Cobalt (HCP) (using FCC shortcut).

$$\rho = \frac{n \cdot A}{V_c \cdot N_A} \quad \left| \quad \begin{array}{l} n=4 \\ A=58.933 \end{array} \right. \quad \left| \quad \begin{array}{l} V_c = (2\sqrt{2}R)^3 \\ R = 200 \times 10^{-12} \text{ m} \end{array} \right.$$

$$\rho = \frac{4 \cdot 58.933}{(2\sqrt{2} \cdot 200 \times 10^{-12})^3 \cdot N_A} = 2.2 \text{ g/cm}^3$$

Another way to do 4, both are fine for density calculations as $APF_{HCP} = APF_{FCC}$:

$$(4b) \quad V_C = \frac{3\sqrt{3}}{2} a^2 \cdot h \quad \left| \begin{array}{l} h = 1.633a \\ a = 2r \end{array} \right.$$

$$\rho = (6 \cdot 58.933) / \left(\frac{3\sqrt{3}}{2} \cdot 1.633 (2 \times 200 \times 10^{-12})^3 \cdot N_A \right)$$

$$= 2.2 \text{ g/cm}^3$$

$$(3) \quad \rho = \frac{n \cdot A}{V_C \cdot N_A} \rightarrow A = \frac{\rho \cdot V_C \cdot N_A}{n}$$

$$A = \frac{9.196 \times 10^6 \cdot (2 \times 168 \times 10^{-12})^3 \cdot N_A}{1}$$

$$= 210.07 \text{ g/mol.}$$

$$(5) \quad \rho = \frac{n \cdot A}{V_C \cdot N_A} = \frac{nA}{(2\sqrt{2}R)^3 \cdot N_A}$$

$$= \frac{(4 \cdot 44.955)}{(2\sqrt{2} \cdot 162 \times 10^{-12})^3 \cdot N_A}$$

$$= 3.1 \text{ g/cm}^3$$

$$\begin{aligned} \textcircled{2} \quad \rho &= \frac{n_a A_a + n_c A_c}{V_c \cdot N_a} \\ &= \frac{n_a A_a + n_c A_c}{(2R_a + 2R_c)^3 \cdot N_a} \end{aligned}$$

$$(2R_a + 2R_c) = \left(\frac{n_a A_a + n_c A_c}{\rho \cdot N_a} \right)^{1/3}$$

$$R_a = \left[\left(\frac{n_a A_a + n_c A_c}{\rho \cdot N_a} \right)^{1/3} - 2R_c \right] \cdot \frac{1}{2}$$

$$\begin{aligned} R_a &= \left[\left(\frac{4 \times 35.453 + 4 \times 107.87}{5.56 \times 10^6 \cdot N_a} \right)^{1/3} - 2.144 \times 10^{-12} \right] \cdot \frac{1}{2} \\ &= 133 \text{ pm} \end{aligned}$$